

A Semi-Analytic Framework for Killed Moments of Leveraged Long–Short Positions under Jump Risk

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Method: survival probabilities and killed moments Application: initial-buffer sizing

DeFi Lending Platforms

Aave / Compound: pooled lending

- ▶ **Mechanism** lenders supply pool assets; borrowers draw pooled liquidity
- ▶ **Accounting** no bilateral lender; protocol-level collateral and debt balances
- ▶ **Interest accrual** variable supply/borrow APRs; utilization-driven index growth
- ▶ **No fixed tenor** open-ended loans; no maturity date; no term structure contract

Overcollateralization

- ▶ Pseudonymous parties; no credit underwriting; no legal margin call
- ▶ Solvency and obligation enforcement entirely collateral-based
- ▶ No-arbitrage discipline through collateral constraint
- ▶ **Liquidator role** repay insolvent debt; seize collateral bonus; restore pool solvency

Health Factor and Boundary

Key mechanism variables

State variables:

$$V_{1,t} := n_1 S_{1,t} \quad \text{collateral value}, \quad V_{2,t} := n_2 S_{2,t} \quad \text{borrow/debt value}$$

Loan-to-value* and liquidation threshold:

$$\text{LTV}_t := \frac{V_{2,t}}{V_{1,t}}, \quad b = \text{threshold LTV}$$

Aave origination $\text{LTV} < \text{threshold LTV } b$; initial buffer by design

Health factor:

$$H_t := \frac{b V_{1,t}}{V_{2,t}} = \frac{b}{\text{LTV}_t}$$

Liquidation triggering condition:

$$H_t < 1 \quad \Longleftrightarrow \quad V_{2,t} > b V_{1,t}$$

Origination requires $\text{LTV}_0 \leq \ell_{\max} < b$, hence $h_0 \geq \log(b/\ell_{\max})$.

Constructing the Long–Short Position

Step	Protocol action	Economic exposure
0	Initial capital injection: \$1-worth BTC	External capital only
1	Supply BTC collateral to Aave	Long BTC collateral
2	Borrow \$0.73-worth ETH	Short ETH liability
3	Sell ETH into \$0.73-worth BTC	Self-financing conversion
4	Re-supply BTC as collateral	Total BTC collateral: \$1.73
Result	Collateralized debt position	Leveraged long BTC, short ETH

Resulting normalized position:

- ▶ External capital: \$1; no additional capital after step 0
- ▶ BTC collateral: \$1.73; ETH debt: \$0.73
- ▶ $(b, \ell_{\max}) = (0.78, 0.73)$; $H_0 = 0.78 \times 1.73/0.73 \approx 1.85$
- ▶ Leveraged BTC exposure; short ETH exposure

Looping the Position

Repeat steps 1–4; zero transaction cost and slippage

With origination ratio $\ell_{\max} < b$, the first m recycled additions are:

$$\ell_{\max}, \quad \ell_{\max}^2, \quad \dots, \quad \ell_{\max}^m$$

Total BTC collateral after m loops:

$$V_1^{(m)} = 1 + \ell_{\max} + \dots + \ell_{\max}^m = \frac{1 - \ell_{\max}^{m+1}}{1 - \ell_{\max}}$$

ETH debt:

$$V_2^{(m)} = \ell_{\max} + \ell_{\max}^2 + \dots + \ell_{\max}^m = V_1^{(m)} - 1$$

Infinite-loop limit:

$$V_1^{(\infty)} = \frac{1}{1 - \ell_{\max}}, \quad V_2^{(\infty)} = \frac{\ell_{\max}}{1 - \ell_{\max}}$$

Aave illustration $(b, \ell_{\max}) = (0.78, 0.73)$:

$$V_1^{(\infty)} = 3.704, \quad V_2^{(\infty)} = 2.704, \quad H_0^{(\infty)} = \frac{b}{\ell_{\max}} \approx 1.068$$

The protocol origination gap prevents entry exactly at the liquidation boundary.

Unwinding and Taking Profit

Favorable scenario: BTC +10%; ETH -16.7%

$$H_t = \frac{0.78 V_{1,t}}{V_{2,t}}, \quad \text{equity} = V_{1,t} - V_{2,t}$$

Scenario	BTC collat.	ETH debt	H	Equity	Action
Entry	\$1.73	\$0.73	1.85	\$1.00	Hold
Favorable move	\$1.903	\$0.608	2.44	\$1.295	Voluntary unwind

Outcome:

$$\text{gain} = 1.295 - 1.000 = 0.295$$

Voluntary unwind; solvent position; profit realization

Aave Liquidation and Immediate Risk

Adverse scenario: BTC -25%; ETH +40%

$$H_t = \frac{bV_{1,t}}{V_{2,t}} = \frac{0.78 V_{1,t}}{V_{2,t}}, \quad H_t < 1 \iff \frac{V_{2,t}}{V_{1,t}} > 0.78$$

Scenario	BTC collat.	ETH debt	H	Equity	Outcome
Entry	\$1.73	\$0.73	1.85	\$1.00	Safe
Adverse move	\$1.298	\$1.022	0.99	\$0.276	Liquidatable

Outcome:

$H_t < 1 \Rightarrow$ keeper liquidation; collateral seized at bonus

Recovery drivers: shock severity; market sentiment/liquidity; liquidator inventory willingness; gas fees

Killed-payoff convention: $\Pi_T = 0$ whenever $\tau \leq T$; this is not a recovery model

Payoff Function from the Trade

Holdings: n_1 units of collateral asset S_1 ; debt: n_2 units of borrowed asset S_2

Killed terminal payoff:

$$\Pi_T := (n_1 S_{1,T} - n_2 S_{2,T}) \mathbf{1}_{\{\tau > T\}}$$

Running BTC/ETH example:

$$S_1 = \text{BTC}, \quad S_2 = \text{ETH}, \quad n_1 S_{1,0} = 1.73, \quad n_2 S_{2,0} = 0.73$$

Therefore:

$$\Pi_T = \left(1.73 \frac{S_{1,T}}{S_{1,0}} - 0.73 \frac{S_{2,T}}{S_{2,0}} \right) \mathbf{1}_{\{\tau > T\}}$$

Survival indicator: $\tau \leq T \Rightarrow \Pi_T = 0$

Initial net book value normalized to one unit of account; moments stated directly in Π_T

Why Payoff Moments Are Hard

Target statistics:

$$\mathbb{E}[\Pi_T], \quad \text{Var}(\Pi_T), \quad \mathbb{E}[\Pi_T^k], \quad k \geq 1$$

Killed payoff:

$$\Pi_T^k = (n_1 S_{1,T} - n_2 S_{2,T})^k \mathbf{1}_{\{\tau > T\}}$$

Terminal payoff state:

$$(S_{1,T}, S_{2,T})$$

Pathwise killing process:

$$\tau > T \iff H_s \geq 1 \text{ for all } s \leq T, \quad H_s = \frac{b n_1 S_{1,s}}{n_2 S_{2,s}}$$

Direct moment:

$$\mathbb{E}_{\mathbb{P}}[\Pi_T^k] = \mathbb{E}_{\mathbb{P}}[(n_1 S_{1,T} - n_2 S_{2,T})^k \mathbf{1}_{\{\tau > T\}}]$$

Difficulty: terminal bivariate payoff + pathwise barrier

Generic numerical routes:

- ▶ Monte Carlo; barrier monitoring / possible 2D Brownian-bridge refinement
- ▶ Numerical PDE/PIDE; two terminal state variables plus killing boundary

Literature Positioning

Intersection:

DeFi liquidation + first-passage methods for jump diffusions

- ▶ **DeFi liquidation** Qin et al. [4]; Heimbach–Huang [1]
- ▶ Incentive structure; leverage amplification; liquidation cascades
- ▶ Empirical focus; no repeated killed-moment evaluation engine
- ▶ **First passage under jumps** Kou–Wang [3]; Sepp [5]
- ▶ Double-exponential jumps; Laplace transforms; finite algebraic systems
- ▶ Classical object: single-asset barrier option

Gap:

barrier-killed long-short ratio $\neq$ single-asset barrier option

Bridge: change of measure $\rightarrow$ one-dimensional killed expectation

Price Process

Jump motivation: crypto gaps; flash crashes; barrier crossed before rebalance

Kou model [2]

Price dynamics, $i = 1, 2$:

$$S_{i,t} = S_{i,0} e^{X_{i,t}}, \quad dX_{i,t} = \mu_i^X dt + \sigma_i dB_{i,t} + d\left(\sum_{n=1}^{N_{i,t}} J_i^{(n)}\right)$$

Brownian correlation:

$$d\langle B_1, B_2 \rangle_t = \rho dt,$$

Poisson intensity: λ_i

Kou double-exponential jump density:

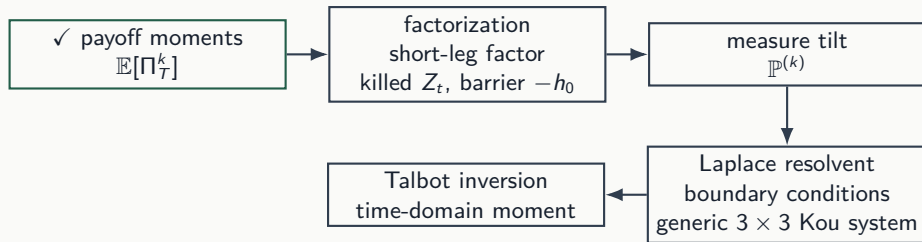
$$f_i(j) = \frac{p_i}{\eta_{i,+}} e^{-j/\eta_{i,+}} \mathbf{1}_{\{j>0\}} + \frac{1-p_i}{\eta_{i,-}} e^{j/\eta_{i,-}} \mathbf{1}_{\{j<0\}}$$

Drift normalization: $\mathbb{E}_{\mathbb{Q}}[S_{i,t}/S_{i,0}] = e^{\mu_i t}$

$$\mu_i^X = \mu_i - \frac{1}{2}\sigma_i^2 - \lambda_i \chi_i, \quad \chi_i = \mathbb{E}[e^{J_i} - 1] = \frac{p_i}{1 - \eta_{i,+}} + \frac{1 - p_i}{1 + \eta_{i,-}} - 1$$

Constant supply/borrow rates absorbed into drift; dynamic rate modeling outside scope

Semi-Analytic Architecture



Covered: payoff function + need for moments

Next: factorization and measure change $\rightarrow$ one-dimensional killed expectation

$$^k \eta_{1,+} < 1, \quad ^k \eta_{2,+} < 1 \quad \text{moment-admissibility condition}$$

Factorization: From Two Assets to One Barrier

Motivation: separate terminal short-leg factor; isolate killed ratio problem

Bivariate payoff; scalar liquidation state:

$$Z_t = X_{1,t} - X_{2,t}$$

Initial log health factor:

$$h_0 := \log H_0$$

Factorization using $H_T = e^{h_0 + Z_T}$:

$$\Pi_T = \frac{n_2}{b} S_{2,0} e^{X_{2,T}} (e^{h_0 + Z_T} - b) \mathbf{1}_{\{\tau > T\}}$$

One-dimensional payoff kernel:

$$g_k(z; h_0) := (e^{h_0 + z} - b)^k,$$

Integer moments:

$$\Pi_T^k = \underbrace{\left(\frac{n_2}{b}\right)^k S_{2,0}^k e^{kX_{2,T}}}_{\mathcal{F}_T^{S_2^2} \text{ part}} \underbrace{g_k(Z_T; h_0) \mathbf{1}_{\{\tau > T\}}}_{\mathcal{F}_T^Z \text{ killed-ratio part}}$$

Measure change: absorb $\mathcal{F}_T^{S_2^2}$ factor; retain killed expectation under Z

Barrier-option analogue: scalar state Z , barrier $-h_0$

Measure Change

Start from factorized moment:

$$\mathbb{E}_{\mathbb{P}}[\Pi_T^k] = \left(\frac{n_2}{b}\right)^k S_{2,0}^k \mathbb{E}_{\mathbb{P}}\left[e^{kX_{2,T}} g_k(Z_T; h_0) \mathbf{1}_{\{\tau > T\}}\right]$$

Exponential tilt:

$$\left. \frac{d\mathbb{P}^{(k)}}{d\mathbb{P}} \right|_{\mathcal{F}_T} = \frac{e^{kX_{2,T}}}{\mathbb{E}[e^{kX_{2,T}}]}$$

Heuristic reminder, with $Y = g_k(Z_T; h_0) \mathbf{1}_{\{\tau > T\}}$:

$$\begin{aligned} \mathbb{E}_{\mathbb{P}^{(k)}}[Y] &= \int Y d\mathbb{P}^{(k)} = \int Y \frac{d\mathbb{P}^{(k)}}{d\mathbb{P}} d\mathbb{P} \\ &= \mathbb{E}_{\mathbb{P}}\left[Y \frac{e^{kX_{2,T}}}{\mathbb{E}_{\mathbb{P}}[e^{kX_{2,T}}]}\right] = \frac{\mathbb{E}_{\mathbb{P}}[e^{kX_{2,T}} Y]}{\mathbb{E}_{\mathbb{P}}[e^{kX_{2,T}}]} \implies \mathbb{E}_{\mathbb{P}}[e^{kX_{2,T}} Y] = \mathbb{E}_{\mathbb{P}}[e^{kX_{2,T}}] \mathbb{E}_{\mathbb{P}^{(k)}}[Y] \end{aligned}$$

Remark: Radon–Nikodym theorem / change-of-measure identity

Reduced moment:

$$\mathbb{E}[\Pi_T^k] = \left(\frac{n_2}{b}\right)^k S_{2,0}^k \mathbb{E}[e^{kX_{2,T}}] \mathbb{E}^{(k)}\left[g_k(Z_T; h_0) \mathbf{1}_{\{\tau > T\}}\right]$$

Bivariate payoff moment $\rightarrow$ killed expectation of the ratio process

Tilted Ratio Process

Remaining object:

$$\mathbb{E}^{(k)} \left[g_k(Z_T; h_0) \mathbf{1}_{\{\tau > T\}} \right]$$

Question:

$$Z_t = X_{1,t} - X_{2,t} \quad \text{under } \mathbb{P}^{(k)}$$

Under $\mathbb{P}^{(k)}$: Z remains a Lévy process

Lévy–Khintchine exponent of Z under $\mathbb{P}^{(k)}$:

$$\psi_Z^{(k)}(s) = \mu_Z^{(k)} s + \frac{1}{2} \sigma_Z^2 s^2 + \lambda_1 (M_1(s) - 1) + \lambda_2 (M_2(k-s) - M_2(k))$$

with

$$M_i(s) = \frac{p_i}{1 - s\eta_{i,+}} + \frac{1 - p_i}{1 + s\eta_{i,-}}, \quad \sigma_Z^2 = \sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2$$

$\mathcal{L}^{(k)}$: infinitesimal generator of Z under $\mathbb{P}^{(k)}$, the generator eigenrelation:

$$\mathcal{L}^{(k)} e^{sz} = \psi_Z^{(k)}(s) e^{sz}$$

Killed Moment PIDE

Target function:

$$U(t, z) := \mathbb{E}_z^{(k)} \left[g_k(Z_t; h_0) \mathbf{1}_{\{\tau(h_0) > t\}} \right], \quad z > -h_0$$

Backward Kolmogorov PIDE for the killed process:

$$\partial_t U(t, z) = \mathcal{L}^{(k)} U(t, z), \quad z > -h_0$$

Initial and absorbing boundary:

$$U(0, z) = g_k(z; h_0), \quad U(t, z) = 0 \quad \text{for } z \leq -h_0$$

One time variable; one state variable z for the ratio process Z

Laplace Transform Trick

Time-domain PIDE:

$$\partial_t U(t, z) = \mathcal{L}^{(k)} U(t, z), \quad U(0, z) = g_k(z; h_0)$$

Laplace transform in time:

$$\widehat{U}(q, z) = \int_0^\infty e^{-qt} U(t, z) dt, \quad q > 0$$

Laplace-domain resolvent:

$$(q - \mathcal{L}^{(k)}) \widehat{U}(q, z) = g_k(z; h_0), \quad z > -h_0$$

Sepp [5] idea:

Laplace transform + double-exponential jumps $\rightarrow$ finite algebraic system

Here:

$$\psi_z^{(k)}(\gamma) = q \implies \text{degree-six characteristic equation}$$

Distinct downward phases: three admissible roots + three boundary conditions $\rightarrow 3 \times 3$ system

Resolvent Solution Form

Resolvent decomposition:

$$\widehat{U}(q, z; h_0) = \widehat{U}_{\text{part}}(q, z; h_0) + \widehat{U}_{\text{hom}}(q, z; h_0)$$

Particular solution:

$$\widehat{U}_{\text{part}}(q, z; h_0) = \sum_{j=0}^k \frac{c_{k,j}(h_0)}{q - \psi_Z^{(k)}(j)} e^{jz}$$

Homogeneous solution:

$$\widehat{U}_{\text{hom}}(q, z; h_0) = \sum_{m=1}^3 B_m^{(k)}(q; h_0) e^{\gamma_{m+3}(q)(z+h_0)}$$

Boundary coefficients:

$$\underbrace{\mathbf{M}_{\text{bar}}^{(k)}(q)}_{3 \times 3} \underbrace{\mathbf{B}^{(k)}}_{3 \times 1} = - \underbrace{\mathbf{b}^{(k)}}_{3 \times 1}$$

PIDE solve $\rightarrow$ summation of exponentials + one generic 3×3 linear system

Primary Output: Survival and Killed Moments

Objective-independent Laplace–resolvent outputs

h_0	p_{surv}	M_1	M_2	$\mathbb{E}[\Pi_T \mid \tau > T]$	$\mathbb{E}[\Pi_T^2 \mid \tau > T]$
0.0662	0.306293	0.385094	0.507211	1.257274	1.655967
0.1000	0.461481	0.536822	0.655481	1.163260	1.420385
0.2500	0.912938	0.898984	0.934991	0.984716	1.024156
0.5000	0.999613	0.984080	1.003811	0.984461	1.004199
1.0000	1.000000	1.015788	1.050402	1.015788	1.050402
2.0000	1.000000	1.033907	1.081910	1.033907	1.081910

$M_k = p_{\text{surv}} \mathbb{E}[\Pi_T^k \mid \tau > T]$; the first row is the protocol origination bound. Decision rules consume this map downstream.

Conditional Statistics and MC Survival

Secondary transformation of killed moments; Laplace–resolvent vs MC survival

h_0	H_0	$p_{\text{surv}}^{\text{Lap}}$	$p_{\text{surv}}^{\text{MC}}$	Mean	Variance	Skewness	Ex. Kurt.
0.066	1.068	0.3063	0.3171	1.2573	0.0752	0.8766	1.0991
0.10	1.11	0.4615	0.4750	1.1633	0.0672	0.8525	1.0083
0.25	1.28	0.9129	0.9156	0.9847	0.0545	0.5676	0.3668
0.50	1.65	0.9996	0.9995	0.9845	0.0350	0.3132	0.3565
0.75	2.12	1.0000	1.0000	1.0043	0.0236	0.3791	0.4093
1.00	2.72	1.0000	1.0000	1.0158	0.0186	0.4158	0.4234
1.50	4.48	1.0000	1.0000	1.0280	0.0145	0.4318	0.4187
2.00	7.39	1.0000	1.0000	1.0339	0.0130	0.4263	0.4017
∞	∞	1.0000	—	1.0414	0.0114	0.4052	0.3612

- ▶ Smaller H_0 : lower survival probability; higher liquidation probability
- ▶ Conditional mean: U-shape across H_0 ; high near boundary, trough around $H_0 \approx 1.3$ – 1.7
- ▶ Conditional variance: decreasing as H_0 increases

Bottom row: no-shorting analytical limit

Numerical Example

Laplace minus MC; wall-clock computation time

h_0	H_0	Δp_{surv}	ΔMean	$\Delta \text{Var.}$	$\Delta \text{Skew.}$	$\Delta \text{Kurt.}$	Lap.	MC
0.066	1.068	-0.0108	0.0040	-0.0019	-0.0236	-0.1965	0.06	48.67
0.10	1.11	-0.0136	0.0042	-0.0013	-0.0230	-0.1499	0.10	49.15
0.25	1.28	-0.0026	-0.0001	-0.0007	-0.0054	-0.0563	0.10	48.90
0.50	1.65	0.0001	-0.0005	-0.0003	-0.0110	-0.0347	0.10	49.11
0.75	2.12	0.0000	-0.0002	-0.0002	-0.0107	-0.0382	0.10	49.49
1.00	2.72	0.0000	-0.0002	-0.0002	-0.0123	-0.0378	0.10	49.46
1.50	4.48	0.0000	-0.0001	-0.0001	-0.0144	-0.0369	0.10	49.94
2.00	7.39	0.0000	0.0000	-0.0001	-0.0152	-0.0362	0.10	49.76

- ▶ $\Delta p_{\text{surv}} < 0$: MC misses intra-period barrier crossings
- ▶ Gap largest near $H_0^{\min}$; negligible for larger health buffers
- ▶ Host-specific runtime: Laplace $\approx 0.10\text{s}$ vs grid-monitored MC $\approx 49\text{--}50\text{s}$

Application: Objective-Specific Sizing

Choice variable: $h_0 = \log H_0$; equivalently initial borrow amount

One illustrative objective:

$$\Phi(h_0) = \mathbb{E}[\Pi_T \mid \tau > T] - \frac{\alpha}{2} \text{Var}(\Pi_T \mid \tau > T) - \delta p_{\text{liq}}(h_0),$$

Liquidation probability:

$$p_{\text{liq}}(h_0) = \mathbb{P}(\tau \leq T)$$

Objective-specific selection:

$$h_0^* \in \arg \max_{h_0 \in \mathcal{H}(b, \ell_{\max})} \Phi(h_0)$$

Alternatives use the same evaluation map: a liquidation constraint or unconditional killed-payoff mean–variance. No rule is designated as universally correct.

Same Engine, Different Selections

Baseline calibration; 200-point grid on $H_0 \in [1.068, 2.000]$

Rule	Parameters	Selected H_0	L_0	p_{liq}
Max. leverage under cap	$\bar{p} = 0.05$	1.331	2.417	0.0478
Max. leverage under cap	$\bar{p} = 0.10$	1.274	2.577	0.0980
Max. leverage under cap	$\bar{p} = 0.20$	1.214	2.799	0.1976
Unconditional killed MV	$\alpha = 1$	$2.000^\dagger$	1.639	$< 10^{-4}$
Conditional MV + penalty	$(\alpha, \delta) = (1, 0)$	$1.068^\dagger$	3.704	0.6937
Conditional MV + penalty	$(\alpha, \delta) = (1, 0.5)$	$2.000^\dagger$	1.639	$< 10^{-4}$

[†] Candidate-grid endpoint. The selected buffer changes with the rule and preference parameters; no row is a universal recommendation.

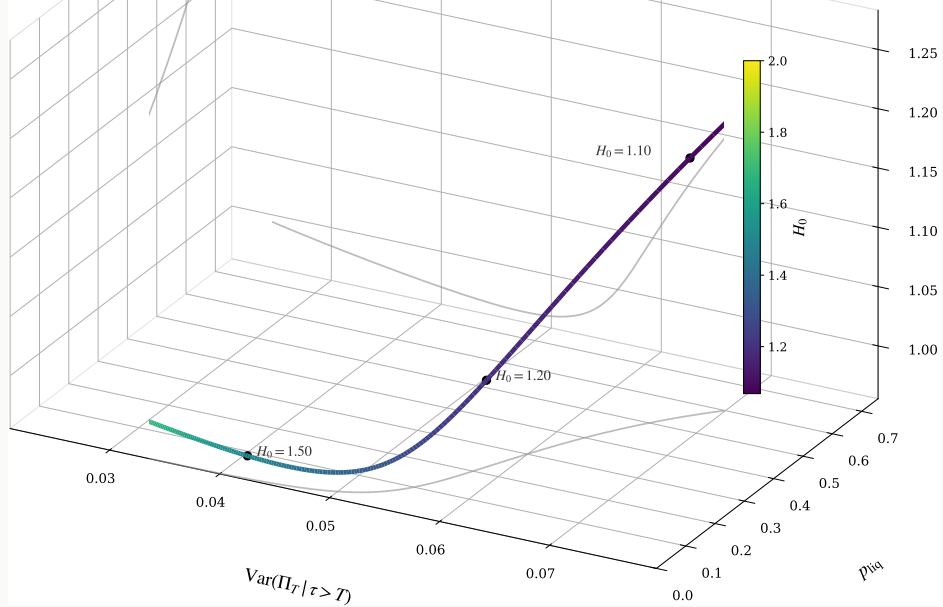
Sampling Uncertainty Is Material

Paired circular moving-block bootstrap: 40 recalibrations, 1.5-day blocks, 90% percentile intervals.
Downstream rule: maximize L_0 subject to $p_{\text{liq}} \leq 0.10$ on the same 200-point grid.

Quantity	Point	90% interval
p_{surv} at $H_0 = 1.274$	0.902	[0.483, 0.978]
Selected H_0	1.274	[1.175, 1.536]
Selected L_0	2.577	[2.032, 2.974]

Conditional on the Kou model, ECF criterion, preprocessing, protocol terms, block length, and sizing grid; not a model-selection or out-of-sample forecast interval.

Efficient frontier, coloured by H_0



Conclusions and Discussion

- ▶ DeFi long-short $\rightarrow$ collateral/debt first passage
- ▶ Measure change $\rightarrow$ one-dimensional killed expectations of $Z = X_1 - X_2$
- ▶ Kou jump diffusion $\rightarrow$ finite-dimensional Laplace resolvent
- ▶ Primary output: $h_0 \mapsto$ survival and killed-payoff moment characteristics
- ▶ Sizing is an objective-specific application of that map

Scope limits

- ▶ Killed-payoff benchmark does not model liquidation recovery
- ▶ Static buffer only: no collateral intervention, oracle/keeper delay, or execution cost
- ▶ Selected buffers remain objective-, model-, and calibration-dependent

Thank you

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